Sara Pidò

PostDoc @ Smart Eyewear Lab (Politecnico di Milano & EssilorLuxottica)

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About Me

AI researcher and **project leader** with a strong background in Machine Learning, Computer Vision, and Natural Language Processing. Experienced in managing multi-track research projects end-to-end, from defining scope and milestones to coordinating cross-functional teams and delivering working systems. My strengths are planning scope and milestones, coordinating teams, and aligning research with industry needs. I am driven by the belief that technology should serve people and improve everyday life. Outside of work, I recharge by running, hiking, cooking, and spending time with people.

Work Experience

Postdoctoral Associate SEL, Milano | 2025 (Ongoing)

- Project lead across 3 concurrent CV & embedded AI projects: define scope, requirements, milestones, and success metrics; keep research, software, and hardware teams aligned end-to-end.
- Lead model development and embedded optimization (quantization, lightweight compression); drive hardware integration from prototype to on-device deployment.
- Supervise and mentor MSc/PhD students on multimodal projects leveraging VLMs and LLMs; manage timelines and deliverables across student teams.

Postdoctoral Associate MIT, Cambridge | 2024

- Project planning**:** Organized workstreams and timelines, ran user studies and supervised students to build explainable AI (XAI) solutions leveraging LLMs.
- Co-designed and developed an anomaly-detection pipeline for Iberdrola, a wind-turbine company.

Data Scientist Intern Calvin Risk, Zurich | May 2023 - August 2023

- Designed, developed, and integrated AI incident-risk models into enterprise risk-management platforms.
- SUpported integration of the EU AI Act for AI governance and responsible AI.

Visiting Research Assistant MIT, Cambridge | 2022

- Led the design and delivery of a time-series pipeline with AutoML.
- Collaborated with Iberdrola on wind-turbine data projects, executing core data-science workflows.

Research Assistant Politecnico di Milano | 2020-2023

- Developed NLP and ML algorithms to automate complex data science processes, working both independently and collaborating within a team, leading master students.

Teaching Assistant Politecnico di Milano | 2020-2022

- Prepared practical lectures on Python programming for the Master's Course in Bioinformatics
- Prepared exercises and help students on C and MATLAB programming for the BSc' students
- Kept a course for Machine Learning for high school students.

Academic History

2020-2023 PhD in Data Analytics and Decision Science, Politecnico di Milano (cum Laude)

Thesis: Exploiting AI and NLP methods for Empowering Naive Users in Solving Data Science Problems

Fellowship offered by ERC Advanced Grant (693174)

2022-2022 Visiting PhD, Data to AI Lab, Massachusetts Institute of Technology

Won fellowship by Rocca Project

2017-2019 M.Sc. Computer Science Engineering, Politecnico di Milano (110/110)

Thesis: Network integration algorithms for the analysis of biological data

2018-2019 Erasmus+ Computer Science, KU Leuven

2014-2017 B.Sc. Computer Engineering, Politecnico di Milano

Thesis: Development of the game Lorenzo il Magnifico

Projects

On-device Computer Vision for Smart Eyewear - Project Lead

Leading end-to-end on-device CV algorithms: define requirements and milestones, align research and development of computer vision algorithms, optimize for embedded constraints (quantization, lightweight compression), coordinate hardware integration, enabling an internal on-device demo and acceptance for pilot. Mentor and direct MSc/PhD students to move from research-only prototypes to on-device deployment.

LLMs for XAI: Future Directions for Explaining Explanations - Project Lead

Investigating the **use of Large Language Models (LLMs) to enhance Explainable AI (XAI)** by transforming ML explanations into human-readable narratives. Focused on refining existing XAI algorithms, resulting in improved interpretability and usability, supported by initial experiments and user studies.

Zephyr: SME Guided Framework for Developing Models for Predictive Maintenance

Collaborating in developing the Zephyr framework to **integrate Subject Matter Experts (SMEs) knowledge** into the ML model development process for predicting wind turbine anomaly detection. Zephyr enables scalable, **data-centric model development** with SME expertise, enhancing both accuracy and viability of predictive models.

Exploiting AI and NLP methods for Empowering Naive Users in Solving Data Science Problems

Making data science accessible to users without data science background by creating new algorithms for the automatic generation, labeling and filtering of **prediction tasks** for **time-series data,** designing new tools for assisting users in developing machine learning pipelines through **natural language processing, conversational agents** and **autoML.**

Computational analysis of fused co-expression networks for the identification of candidate cancer gene biomarkers

Developing a general framework to **infer relevant gene biomarkers** using multiple **gene co-expression networks** for each cancer type, specifically, of genes from kidney renal clear cell carcinoma, liver hepatocellular carcinoma, and prostate adenocarcinoma data sets of **TCGA database**. The gene communities are extracted through a **data-driven pipeline** and then evaluated through both functional analyses and literature findings.

Publications

Latest publications:

- Corti G., Veronesi C., Bartoli P., **Pidò S.,** Giudici A., Palermo F., Trojaniello D., Matteucci M., Mentasti S. Comparative Analysis of Lightweight Vision Models on Embedded Accelerator Devices. ICIAP 2025 (Accepted).
- Zytek A,, **Pidò S.,** Alnegheimish S., Berti-Equille L., Veeramachaneni K. Explingo: Explaining AI Predictions using Large Language Models. IEEE Big Data 2024.
- Zytek, A.*, **Pidò S.***, and Veeramachaneni, K.. LLMs for XAI: Future Directions for Explaining Explanations. HXCAI, CHI2024.
- **Pidò S.,** Pinoli P., Crovari P., Ieva F., Garzotto F. and Ceri, S. Ask Your Data—Supporting Data Science Processes by Combining AutoML and Conversational Interfaces. IEEE Access.

Check out my Google Scholar profile for more: https://scholar.google.com/citations?user=M_SrptEAAAAJ&hl=en

Skills

Management: PRDs, roadmaps, prioritization, KPI design, stakeholder management, defining milestones, dependencies, cross-team coordination, user research, experiment design.

Data/AI:

- Machine Learning, Natural Language Processing (LLMs), Deep Learning fundamentals
- Data analysis and visualization
- Python (Pytorch, Tensorflow, Keras, Scikit-learn, Matplotlib, Seaborn), Matlab, C.
- Google Cloud, AWS, Azure.

Tools: Confluence, Notion, GitHub, SQL, Office.

Languages: Italian (native), English (C1)

Referees

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